Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/03/2018
PI Name:	Goree, John Ph.D.		
Project Title:	Nonlinear Wave Experiments in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	06/14/2017	**End Date:**	06/13/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: Change in start and end dates to 6/14/2017-6/13/2021 (from 4/15/2017-4/15/2021) per I. Hahn/JPL and PI (Ed., 1/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Bin Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1573629		
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Performance Goal Text:			

Task Description:

A flight experiment will be performed and analyzed in a study of nonlinear synchronization of waves. Synchronization is a cooperative phenomenon for oscillators or waves. A signature of synchronization is a frequency that shifts when an external modulation is applied. Compared to point-like oscillators, waves are more complicated due to their spatio-temporal behavior and dispersion properties. An understanding of synchronization is lacking for waves of all kinds, including those in dusty plasmas.
The experiment will use dusty plasmas, also called "complex plasmas." These are mixtures of small particles of solid matter, electrons, ions, and neutral gas. The small solid particles obtain large electric charges, so that they exhibit collective behavior. They are electrically confined, so that they do not contact any boundaries.

Microgravity enables this experiment by suppressing sedimentation, thereby allowing the waves to propagate distances of many wavelengths, with relatively uniform background conditions. This experiment is based on previous ground-based experiments, where waves propagated only a few wavelengths.

Our flight experiment will be performed using PK-4, which is a facility in the Columbus Module of the International Space Station (ISS), built by the European Space Agency.

SCIENTIFIC/TECHNICAL MERITS

The main scientific topic is the synchronization of a dust acoustic wave (DAW). A DAW is a low-frequency compressional wave, which involves charge-separation from electrons and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Synchronization experiments are performed by applying electrical waveforms to an electrode to modulate the dusty plasma sinusoidally. The diagnostic for detecting the wave motion is video imaging using particle observation cameras. Image data will be analyzed to prepare space-time diagrams, which provide a spatial and temporal description of the wave's amplitude as it propagates. These space-time diagrams are the foundation for further analysis methods.

In support of our experiment, a theoretical effort will be undertaken at Iowa. The goal is to contribute to the development of a theory for synchronization of propagating waves, to compare to the experiment. The theoretical effort at Iowa will focus on numerical simulations using a hybrid model. Molecular dynamics simulation will also be performed. The main theoretical effort, which will center on analytic theory, will be done by theorist elsewhere, and we will collaborate with them.

The feasibility of our flight experiment is based on our previous ground-based experiments, which confirm that synchronization of the DAW occurs. These ground-based experiments also revealed unexplained phenomena, including a nonharmonic synchronization state, which we will investigate with our flight data.

In the unlikely event that our main experiment with waves does not yield useful results, we have a plan for a backup experiment to investigate mobility-limited motion of particles having a different size.

BROADER IMPACTS

This research has an overlap with nonlinear dynamics as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, a website, and Twitter. A K-12 outreach presentation will be made at an elementary school.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

During this period, our group carried out ground-based research to analyze flight data and to prepare for forthcoming flight experiments using the PK-4 instrument on the International Space Station (ISS). The research activities included flight experiment planning, analysis method development, and theoretical support. This research has led to five peer-reviewed journal papers including the one published on Nature Physics, two invited and seven contributed talks, and one PhD thesis.
This research focused on the fundamental physics of dusty plasma. Small particles of solid matter, also called dust, are immersed in a mixture of electrons, ions, and neutral gas, making a dusty plasma. These small particles obtain large electric charges, so that they exhibit collective behavior like waves and viscous transport. In this project, we investigated the fundamental nature of a nonlinear wave called dust acoustic wave (DAW).

NONLINEAR DUST ACOUSTIC WAVES

A DAW is a low-frequency compressional wave, which involves charge-separation among the dust, electrons, and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Two nonlinear effects were studied: nonlinear waveforms and nonlinear synchronization. A signature of synchronization is a frequency shift due to an external modulation.

FLIGHT EXPERIMENT PLANNING

Our planning involved a collaboration with other institutions or agencies. Significant collaborators included PK-4 Core Team members in Germany and Russia, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, and the collaborators specific to Iowa projects. Both Principal Investigator (PI) John Goree and Co-Investigator Bin Liu interacted with these collaborators frequently via emails and occasionally by phone, and we also met them at various meetings and conferences. One significant meeting was the recent Facility Science Team (FST) meeting at the DLR (German Space Agency) in December 2017. Both PI John Goree and Co-I Bin Liu attended this meeting.

Working with our collaborators, we made a general plan for our forthcoming Iowa experiments on nonlinear wave using the PK-4 instrument onboard the ISS. A preliminary experiment is tentatively planned for late 2018, followed by another experiment about a year later.

To prepare for the forthcoming flight experiment, we performed an on-site laboratory test at the DLR. We did this test together with PK-4 Core Team scientists at the DLR, using a PK-4 Ground Model (the apparatus was the Scientific Reference Model at the DLR). The outcome of this test was a first draft of the flow charts, which will be used to develop more detailed scripts for the upcoming Iowa flight experiments.

Task Progress:	Planning activities on-site at Iowa included a survey of the flight data from previous missions. During the past year, the Iowa group received the flight data from four previous missions, and we have so far surveyed the data for three missions. To expedite our data analysis, we also hired an undergraduate, who was funded by National Science Foundation (NSF). The flight data survey has so far resulted in two significant validations: 1) Can waves be self-excited in PK-4 onboard the ISS? We confirmed that waves can be self-excited in PK-4. This excitation naturally occurs in dust clouds trapped either by radio-frequency or DC powered discharge plasmas. 2) Can the PK-4 plasma under microgravity conditions sustain wave propagation for many wavelengths? We confirmed that microgravity conditions enabled the containment of a much longer 3D dust cloud than was possible under 1g conditions. In one experiment, we found that waves could be seen propagating for more than 30 wavelengths, as compared to only a few wavelengths under 1g conditions. In addition to the two significant validations above, progress has been made in two areas, in preparing our future flight experiment. One area of progress was the refinement of a new scientific topic, which we have already proposed in the recent FST meeting at the DLR. This topic involves a characterization of the waveforms of nonlinear waves. We will compare the experiment to a fundamental theory for dusty plasma, developed by our collaborators in India and the Naval Research Laboratory (NRL). Another area of progress was the development of software for our flight data analysis. Using this software, we successfully extracted experimental parameters from PK-4 raw data products. This progress is significant because PK-4 raw housekeeping data products are not formatted in a way that is understandable to scientists. Therefore, we have developed our own computer code to extract the experimental parameters, and we have shared this software with other NASA-funded PIs. Furthermore, our data survey helped us in optimizing flight experiment conditions. Using an analysis method called a “space-time diagram,” we identified the gas pressure range where plasma glow exhibits a significant disturbance called a striation. We expect that striations will be a common feature that will often affect PK-4 experiments. We also found a limitation in the early PK-4 flight data, due to the choice of camera-frame-rate parameters. This helped us in better defining our experimental requirements. ANALYSIS METHOD DEVELOPMENT To develop the analysis method to be used with our future flight experiment, we analyzed two previous breadboard experiments, involving nonlinear waves and the synchronization of waves. For the nonlinear waves, our effort focused on the waveforms of the waves. We characterized the time variation of the waves, as observed from the ground-based experiments, and we investigated whether the nonlinearities can be quantified using a single parameter. For the synchronization, we adapted an earlier analysis method (validated previously in our earlier ground-based experiments) for use in our future analysis of flight-experiment data. A significant finding is that we confirmed that we can characterize the degree of nonlinearity using a single quantitative measure. To do this, we first checked that the waveforms of the nonlinear DAWs can be accurately described by a Cnodial wave theory. This theory has a fit parameter to characterize the shape of the waveforms that we can use as a measure of nonlinearity. THEORETICAL SUPPORT In support of our experiment, a theoretical effort is underway with collaborators. The goal is to develop a theory for nonlinear waves, to compare to the flight experiment. In June 2017, PI John Goree organized the first team meeting of the theoretical support at the University of Iowa. He arranged for the visit of our collaborators from the Institute for Plasma Research, India, and the Naval Research Laboratory (NRL). Other participants to this meeting included Co-I Bin Liu and a postdoc at the University of Iowa. One outcome was the development of theoretical solutions useful for direct comparison to the experimental data for nonlinear dust acoustic waves. BROADER IMPACTS A K-12 outreach presentation was made at an elementary school. Our research results have been disseminated in journals and at conferences. Five papers have been published in peer-reviewed journals. Two invited and seven contributed talks were presented at international conferences. We also distributed to other PIs a copy of software that we developed at Iowa.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
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